

Development of a Hybrid AES–ECC–Post-Quantum Cryptographic Model for Future-Proof Secure Communications

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Abstract— In this paper, the contribution has been the development of a hybrid AES-ECC-Post quantum cryptography model that would bring about future-proof secure communications in response to the threat of changing and evolving classical and quantum computing. Classical cryptographic systems based on either symmetric or asymmetric systems are limited in nature regarding the management of keys, computational costs, and long-term security, especially in the face of the perceived effects of quantum attacks on commonly used public-key algorithms. In order to find solutions to these issues, the proposed model will combine the Advanced Encryption Standard (AES) to encrypt and decrypt data efficiently and fast, Elliptic Curve Cryptography (ECC) to perform lightweight authentication and compatibility, and the post-quantum cryptography primitives to establish quantum-resistant keys. The hybrid architecture will be interoperable and modular to facilitate an easy way of integrating with other existing communication infrastructures and making the migration towards post-quantum preparedness possible. Performance and security analysis show that the given solution presents a reasonable trade-off between efficiency, scalability, and cryptographic strength in the environments of clouds, enterprise, and Internet of Things (IoT). This paper will integrate classical and post-quantum methods into a single clear framework, hence, offering a deployable and practical solution to the next-generation secure communication systems.

Keywords: Hybrid Cryptography, Post-Quantum Cryptography, AES, Elliptic Curve Cryptography, Quantum-Resilient Security, Secure Communication Systems, Interoperable Cryptographic Models

I. INTRODUCTION

The need to secure communication has been a key requirement of digital systems, and it has emerged as the major mechanism of safeguarding sensitive information, protecting privacy, and building trust throughout communication networks [1]. Hybrid cryptographic frameworks have also become the focus of increasing interest in recent years, and they are necessary to improve interoperability and security in contemporary communication

infrastructures [2]. As people increasingly integrate smart consumer electronics and interconnected digital systems, cryptography measures are essential in ensuring that the data is not easily misused (Hans and Marty 268) [3]. Nonetheless, the advent of quantum computing has brought to the fore specific foul play with the security of traditional cryptographic systems over the long term, especially in sensitive communications like military and defense communication [4]. This has led to an alarming situation where cryptographic solutions are urgently needed and resistant to both classical and quantum-based attacks [5].

With the ever-growing and global nature, the need for secure, efficient, and scalable cryptographic mechanisms has increased significantly. Conventional cryptographic designs tend to be weak in terms of either performance or security in large-scale and real-time configurations [6]. The hardware-based cryptographic models have been suggested to enhance efficiency and reliability, and it is proven that an optimized cryptographic architecture is required to enhance secure communication [7]. Meanwhile, network-level security tools have to consider performance and compatibility with protocols to guarantee the successful transmission of data between different systems of heterogeneous nature [8].

Hybrid cryptography is one of the most popular methodologies for addressing these issues, as it is possible to use the benefits of both symmetric and asymmetric algorithms. It has been demonstrated that hybrid AES-ECC models have the ability to enhance both the security and the efficiency of data assurance through the combination of rapid symmetric encryption and safe asymmetric key administration [9]. Further extensions of the hybrid technique have been made with advanced cryptography applications supporting secure multimedia communication and intelligent system integration [10]. The effectiveness of hybrid models in practical security implementations has been shown by these techniques.

However, the classical hybrid cryptographic constructions can be attacked by quantum attacks, which have prompted the research of more advanced cryptographic primitives. Safe transaction system frameworks and the cryptography performance analysis have highlighted the drawbacks of current algorithms in their implementation to the new generation communication systems [11], [12]. Besides, the cryptographic schemes that are enabled by authentication have also underscored the importance of more robust identity verification schemes in secure communication environments [13].

In an effort to overcome these drawbacks, recent studies have worked on the assessment and improvement of hybrid cryptographic systems in cloud-based and distributed systems [14]. Moreover, post-quantum cryptography systems, including cryptography that is based on NTRU, have been shown to show considerable promise in ensuring that future communication systems will remain inaccessible to quantum attackers [15]. Inspired by these, this paper proposes how a hybrid AES-ECC-Post-Quantum model of cryptograph model can be developed to facilitate effective, interoperable, and future-proof secure communication systems.

II. WORKFLOW OF THE PROPOSED MODEL

The process flow of the suggested hybrid model of AES-ECC-Post-Quantum cryptography, as shown in Figure 1, is a structured and ordered process flow aimed at offering secure, efficient, and quantum-resistant communication. Every step of the workflow is well structured to cover certain security goals, including authentication, key setup, confidentiality, integrity, and future quantum attack resistance.

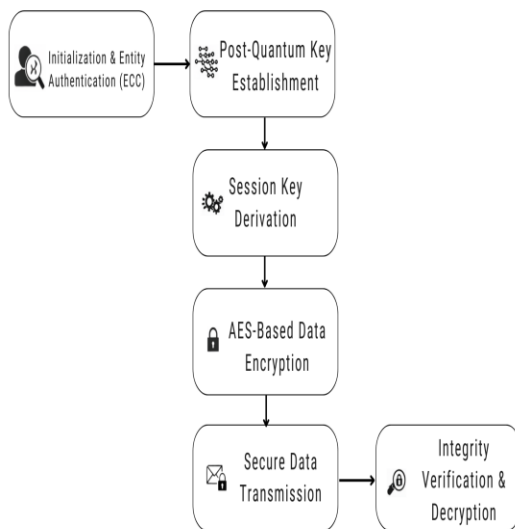


Figure 1: Workflow of the proposed hybrid AES–ECC–Post-Quantum cryptographic model.

The model offers the means of ensuring long-lasting security and short-term practicality by integrating the use of classical and post-quantum cryptography approaches. The first phase of the workflow is the Initiation and Authentication of an entity stage, wherein the party involved in communicated forms of trust through Elliptic Curve Cryptography (ECC) with the other parties.

The reason ECC is selected is that it provides strong authentication with a relatively small key size, which is, correspondingly, in turn, more inclined to a trade-off of the perceived cost of computation compared to traditional asymmetric algorithms. This step does not allow an unauthorized party entry into the communication process and forms a safe base on which future cryptography processes will take place. Once it has been successfully authenticated, the system changes to the Post-Quantum Key Establishment stage. In this stage, a key exchange system resistant to quantum is used to create a shared secret between the communicating parties. In contrast to the classical alternatives involving key exchange, post-quantum approaches are ones that are secure despite having a powerful quantum computer. The step is essential in ensuring the future-proofing of the cryptography framework against potential attacks by quantum devices. The produced shared secret is used in the next stage, the Session Key Derivation stage. A secure key derivation function is applied to produce session-dependent symmetric keys of the shared secret. These keys are temporary in nature in that they are only used during one cycle of communication, thus limiting the attack on key compromise and improving overall security. Upon obtaining the session key, the next step of the workflow is the AES-Based Data Encryption. The plaintext data is being encrypted using the Advanced Encryption Standard (AES), as it is very efficient and has low latency alongside highly tested robustness in data encryption. Through the use of symmetric encryption to provide confidentiality to data, the system has the benefit of ensuring high performance at a high level of security. The secure data is redirected in the communication channel after encryption in the Secure Data Transmission stage.

At this step, encrypted information is transmitted via possibly insecure networks without leaking confidential information. Coded communication is confidential even in an open system or a hostile system. The final stage is the Integrity Verification and Decryption of the message at the receiving end. The integrity verification mechanisms are first implemented to ensure that the data has not been distorted in the course of transmission. After verification of the integrity of the data, the encrypted data is decrypted under the same AES session key that is used to retrieve the original plaintext. This is the final stage in the secure provision of communication, and provides the assurance of the authentic data and integrity of the received data. In general, the proposed cryptographic model can be evaluated as effective due to the combination of three approaches to the workflow, which is shown in Fig. 1,

including ECC-based authentication, post-quantum key establishment, and AES-based data encryption, to develop a balanced solution that provides efficiency, interoperability, and quantum resilience. The given workflow structure enables the use of the proposed model in cloud solutions, enterprise networks, and the Internet of Things (IoT), where long-term security is an essential factor.

III. PROPOSED MODEL

The hybrid AES-ECC-Post-Quantum cryptography model is created to respond to the increasing security gaps in contemporary communication systems due to conventional and novel quantum attacks. As computational technologies rapidly develop, the usual cryptographic mechanisms are becoming more and more susceptible to attacks that can breach the confidentiality of data over a period. In contrast to more traditional cryptographic schemes, which make use of only one type of algorithmic method, the given model incorporates symmetrical, asymmetric, and post-quantum cryptographic schemes in the form of a ranked and harmonious system.

As shown in Figure. 2, the model will be developed in such a way that it will provide a safe entity authentication, a strong key establishment, effective data encryption, and good integrity checking, while being compatible and interoperable with the current cryptographic infrastructures.

The general architecture can securely communicate between a sender and a receiver based on a layered security mechanism, with each layer taking a specific cryptographic task. This division of duties eases separation of duties, fault isolation, and increases maintainability. Additionally, the individual cryptographic elements can be easily upgraded and even replaced, and hence the modular structure provides this dynamic later in life with the advance of cryptography standards, particularly due to the ongoing advancement of post-quantum cryptography. Due to this, the proposed model offers both short-term security advantage and long-term flexibility

A. Major Management and Hybrid Algorithms Coordination.

The Key Management module is the main control and coordination point of the proposed cryptographic model. It manages cryptography parameters, public and private keys, and choice of algorithms, both Post-Quantum Cryptography (PQC) and Elliptic Curve Cryptography (ECC), as shown in Figure 2. The model not only provides compatibility with all available traditional key exchange algorithms, but it also supports post-quantum-resistant key exchange algorithms, thus providing backward compatibility with the existing security systems. Moreover, there is the key management module that manages all the key life cycle, such as key

generation, secure distribution, controlled storage, periodic reenactment, and key revocation. Such an extensive key management system eliminates the chances of key compromise and provides security in the environment under which cryptographic content is handled in a communication process.

B. Entity Authentication and ECC Initialization.

At the stage of initialisation, the Elliptic Curve Cryptography (ECC) is used to perform entity authentication. The reason why ECC is chosen is that it can offer high levels of cryptographic security with much reduced key sizes than traditional asymmetric algorithms, as shown in Figure 2. It leads to decreased computation cost, decreased memory usage, and accelerated authentication procedures, and thus ECC is especially applicable in environments with constrained resources (IoT devices and mobile systems).

This is an authentication step wherein a mutual trust is created between the sender and recipient as they verify each other, and only after that, an exchange of sensitive data will take place. Authenticated correctly will work well to prevent illegal access, impersonation, and replay attacks, and thus provide a secure basis upon which further cryptographic operations can be based.

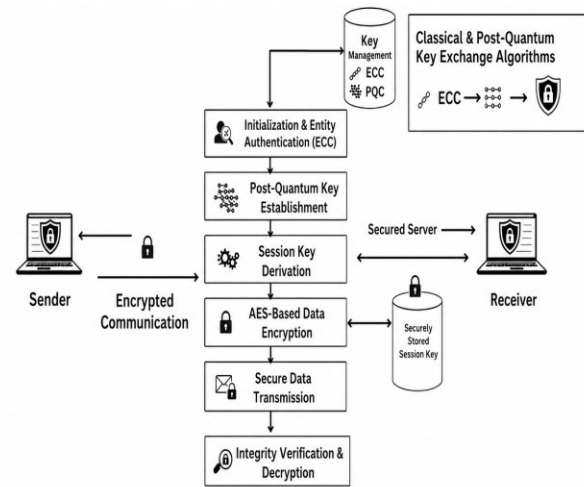


Figure 2: End-to-end secure communication architecture of the proposed hybrid AES-ECC-Post-Quantum model between sender and receiver.

C. Post-Quantum Key Establishment Mechanism.

After the process of authentication is successfully completed, the system initiates the Post-Quantum Key Establishment process. Under this step, a key exchange algorithm, which is quantum-resistant, is used to create a shared secret between the communicating parties, as shown in Figure 2. Post-quantum algorithms are strengthened to ensure that their security is not compromised even when faced with a large and

strong quantum computer that can execute quantum cryptanalysis attacks. The current capabilities of the proposed model are its ability to incorporate post-quantum techniques by this step to maintain cryptographic resistance over time, as well as secure communication sessions against future cryptanalytic victories. This is a design option that is highly relevant to those applications where data confidentiality should be maintained over a long period of time.

D. Derivation and storage of Secure session key.

The secret generated in the post-quantum key establishment phase is handled in the Stage of Session Key Derivation, as shown in Figure 2. The symmetric session keys are uniquely generated with a secure key derivation function that belongs to a communication session. These session keys are short-lived and can only work within the duration of the session, thereby minimizing the threat of key reuse.

Based on the proposed model, the resulting session keys can be stored safely to be used under controlled conditions in current communication. After the session is completed, the keys are thrown away, hence restricting the effect of possible key exposure and maximizing general system security.

E. High-performance Data encryption using E. AES.

The plain-text data is encrypted with the derived session key in the stage of AES-Based Data Encryption. Advanced Encryption Standard (AES) is chosen because of its encrypting speed, low latency, and it has been tested to be resistant to cryptanalytic attacks, as shown in Figure 2. AES is a commonly accepted secure symmetric encryption standard, and AES can efficiently handle huge amounts of data. The proposed model can use AES to encrypt bulk data and provide high performance and good guarantees of confidentiality. This renders the system applicable to high-throughput usage like cloud service systems, enterprise data transfer systems, and real-time communication systems.

F. Secure Data Transmission

In the Encryption phase, the safe data is sent via the communication channel after the encryption process is completed. The data is encrypted, which means that it is protected against the attacks of interception, eavesdropping, and traffic analysis since the data cannot be detected, as shown in Figure 2. This stage manages confidentiality as well as through secure means of delivering information over the untrusted networks that encompass both public and shared communication infrastructure, such as the internet.

G. Integrity Verification and Decryption

Integrity Verification and Secure Decryption stage at the receiver end ensures the data is authentic and quite correct.

Mechanisms of integrity verification are used to ascertain that the data received has not been changed in the processes of transmission, as shown in Figure 2. Any changing or altering opportunities might be identified at this point. After integrity has been checked successfully, the encrypted information is then decrypted with the help of the relevant AES session key in order to obtain the original plaintext. This last step ensures that the information received by the receiver is correct, authentic, and unaltered.

H. Security and Interoperability Benefits of the Proposed Model.

The presented hybrid cryptographic model has a number of significant benefits over the conventional cryptographic methods, as shown in Figure 2. Based on imitation of ECC-based authentication, post-quantum key establishment, and AES-based encryption, the model generates an optimum trade-off between performance and security and future security preparedness. It is composed of modules and layers, which will ensure a seamless interoperability with the existing cryptographic systems and mechanisms towards a gradual shift to post-quantum security. As a result, the proposed model can be easily applied in an extremely diverse variety of application fields, such as cloud computing, enterprise networks, military communications, and Internet of Things (IoT) networks, where efficiency in the short term and long-term security are critical concerns.

IV. RESULT ANALYSIS

The AES-ECC-Post-Quantum hybrid cryptographic model proposed is analyzed in the form of its results by using a prototype secure communications application. The sender-side algorithm, as shown in Figure 3, indicates that plaintext data were encrypted and transmitted successfully and safely. The shown status proves that the authentication, key establishment, and AES-based encryption operations are done properly, which confirms the functionality of the sender module.

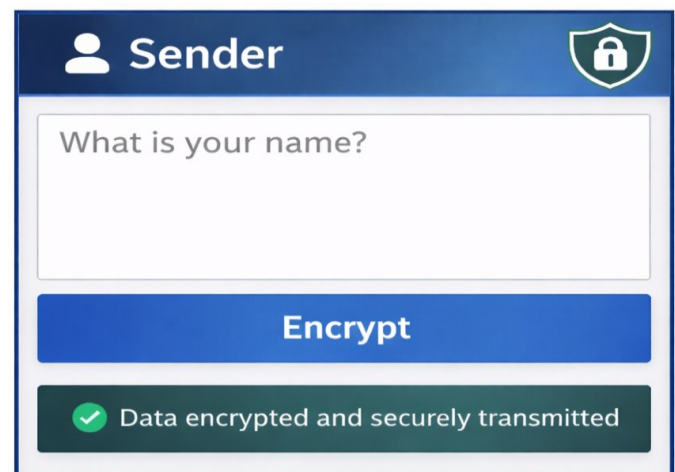


Figure 3: Sender-side user interface showing plaintext input and successful encryption using the proposed hybrid cryptographic model.

The operation on the receiver-side is illustrated in Figure 4, where the encrypted data sent by the sender is successfully decrypted and integrity checked. The output verifies the fact that the obtained session key is used properly when decryption is performed and that the integrity check system detects no alterations to the data sent. This justifies the accuracy and consistency of the decryption and integrity check processes of the intended model.

The efficiency of the proposed cryptographic model in its performance is determined by comparing it with the classical cryptographic algorithms, as shown in Figure 5. The graph shows the change in the processing time as the number of transmissions increases. It is clearly seen that the proposed hybrid model has a low processing time as compared to the classical approach. This is credited to minimal post-quantum operations used in key establishment and the use of effective AES-based symmetric encryption to transmit data, and this leads to improved scalability and low computational overhead.

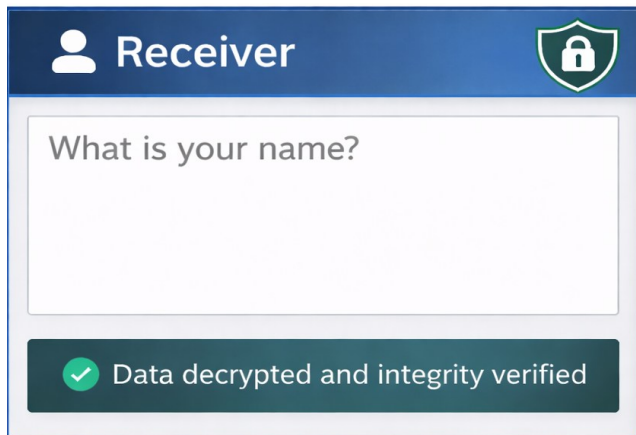


Figure 4: Receiver-side interface for secure data decryption and verification.

On the whole, the findings received on the basis of application-based testing and performance analysis prove that the offered hybrid cryptographic model provides the functionality of secure encryption-decryption of messages, the integrity control that can be carried out successfully, and the enhancement of the efficiency of performance. These results indicate the appropriateness of the proposed solution to the current and future safe communication systems that need not only to be efficient but also to be quantum-resistant.

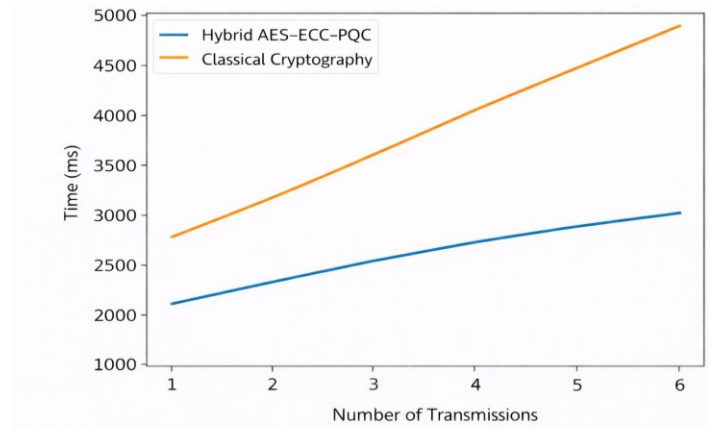


Figure 5: Variation of cryptographic processing time with respect to the number of transmissions for hybrid and classical cryptographic approaches.

V. CONCLUSION

This paper has discussed a hybrid AES-ECC-Post-Quantum cryptography model that is proposed to offer secure, efficient, and futureproof communication with both classical and quantum threats. The proposed framework includes ECC-based authentication, post-quantum key establishment, and AES-based symmetric encryption, with the integration of the three techniques, thus overcoming the constraints of the classical cryptographic schemes but still being compatible with the existing infrastructure. The prototype safe communication application evaluation proved the rightness of the encryption, decryption, and integrity verification procedure, whereas the performance analysis showed that the hybrid model takes lower processing time and has higher scalability than classical cryptographic algorithms. These findings suggest that post-quantum security integration can have no major effect on system efficiency with the incorporation of high-performance symmetric encryption. In general, the suggested model is a reasonable and balanced proposal to next-generation secure communication systems, and it is applicable to the next generation of cloud computing, enterprise, military communications, and Internet of Things (IoT) systems that need long-term cryptographic resilience.

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